Social Choice and Welfare

1 Social Choice and Arrow's Theorem

1. Positive vs. Normative Economics.

2. A social choice problem arises whenever any group of individuals must make a collective choice among a set of alternatives.

3. How can we go from the often divergent, but individually consistent, personal views of society's members to a single and consistent social view? How to aggregate individual preferences into social preferences?

4. A Social Preference Relation (R): a complete and transitive binary relation on the set X of social states. For $x, y \in X$, xRy means "x is socially at least as good as y". Let P and I be the associated relations of strict social preference and social indifference, respectively.

5. Condorcet's Paradox illustrates that the majority voting can fail to satisfy the transitivity requirement on R.

6.

person1	person 2	Person 3
x	y	z
y	z	x
z	x	y

7. How can we go from consistent individual views to a social views that is consistent (both complete and transitive) and that also respects certain basic values shared by members of the community?

8. Social welfare function f, where

$$R = f(R^1, R^2, ..., R^N)$$

That is, f takes N-tuple of individual preference relations on X and maps them into a social preference relation on X.

9. Arrow's requirement on social welfare functions:

   (a) Unrestricted Domain (U). The domain must include all possible combinations of individual preference relations on X (assuming X is finite here).

   (b) Weak Pareto Principle (WP). For any pair of alternatives $x, y \in X$, if xP^iy for all i, then xPy.

   (c) Independence of Irrelevant Alternatives (IIA). Let $R = f(R^1, ..., R^N)$, $\widetilde{R} = f(\widetilde{R}^1, ..., \widetilde{R}^N)$, and for any $x, y \in X$, if each individual i ranks x versus y under R^i the same way that he does under $\widetilde{R}^i$, then the social ranking of x versus y is the same under R and $\widetilde{R}$.

   (d) Non-dictatorship (D). There is no individual i such that for any $x, y \in X$, xP^iy implies xPy regardless of the preferences R^j of all other individuals $j \neq i$.

10. Arrow's Impossibility Theorem. If there are at least three social states in X, then there is no social welfare functions f that simultaneously satisfy conditions U, WP, IIA and D.

11. Proof. Step 1. Consider any social state, c. Suppose each individual places c at the bottom of this ranking. By WP, the social ranking must place c at the bottom as well.

12. Step 2. Imagine now moving c to the top of individual 1's ranking, leaving of all other states unchanged. Next, do the same with individual 2. Continue doing this one individual at a time, keeping in mind that each of these changes in individual preferences might have an effect on the social ranking. Eventually, c will be at the top of every individual's ranking, and so it must then also be at the top of the social ranking by WP. Consequently, there must be a first time during this process that the social ranking of c increases. Let individual n be the first such that raising c to the top of his ranking causes the social ranking of c to increase.

13. A Claim. When c moves to the top of individual n's ranking, the social ranking of c not only increases but c also moves to the top of the social ranking. Suppose not. That is, αRc and $cR\beta$ for some states $\alpha, \beta \neq c$. Now because c is either at the bottom or at the top of each individual's ranking, we can change each individual i's

preference so that $\beta P^i \alpha$, while leaving the position of c unchanged for that individual. By WP $\beta P^i \alpha$ for every individual implies that $\beta P \alpha$. On the other hand, however, because the rankings of c relative to α and of c relative to β have not changed in any individual's ranking, by IIA and transitivity we have $\alpha R \beta$, which contradicts $\beta P \alpha$.

14. Step 3. Consider now any two distinct social states $a, b \neq c$. Change the profile of preferences as follows: change individual n's ranking so that $aP^n cP^n b$, and for every other individual rank a and b in any way so long as the position of c is unchanged for that individual. Note that in the new profile of preferences the ranking of a to c is the same for every individual as it was just before raising c to the top of individual n's ranking in step 2. Therefore, by IIA, the social ranking of a and c must be the same as it was at the moment. But this means that aPc because at that moment c was still at the bottom of the social ranking.

15. Similarly, in the new profile of preferences, the ranking of c to b is the same for every individual as it was just after raising c to the top of individual n's ranking in step 2. Therefore by IIA, the social ranking of c and b must be the same as it was at the moment. But this means that cPb because at that moment c had just risen to the top of the social ranking.

16. Because aPc and cPb, we may conclude by transitivity that aPb. Note that no matter how the others rank a and b, the social ranking agrees with individual n's ranking. By IIA, and because a and b were arbitrary, we may therefore conclude that for all states a and c distinct from c

    $$aP^n b \text{ implies } aPb.$$

    That is, individual n is a dictator on all pairs of social states not involving c.

17. Step 4. Let a be distinct from c. We may repeat the above steps with a playing the role of c to conclude that some individual is a dictator on all pairs not involving a. However, recall that individual n's ranking of c (bottom or top) affects the social ranking of c (bottom or top). Hence, it must be individual n who is the dictator on all pairs not involving a. Because a was an arbitrary state distinct from c, and together with our previous conclusion about individual n, this implies that n is a dictator.

18. The strategy of the proof is to show that conditions U, WP, and IIA imply the existence of a dictator. Consequently, if U, WP, and IIA hold, then D must fail to hold, and so no social welfare function can satisfy all our conditions.

2 Measurability, Comparability, and Some Possibilities

1. How to rescue social welfare analysis from the Arrow's impossibility theorem? One way is to relax the requirements that must be satisfied by the social relation R. For example, replacing transitivity by a weaker condition called "acyclicity", and replacing the requirement that R order all alternatives from best to worst with the simple restriction that we be merely capable of finding a best alternative among any subset, opens the way to several possible choice mechanism, each respecting the rest of Arrow's conditions.

2. Similarly, if transitivity is retained, but condition U is replaced with the assumption that individual preferences are single-peaked, Black (1948) has shown that majority voting satisfies the rest of Arrow's conditions, provided that the number of individual is odd.

3. Another approach is about the interpersonal comparison of individual preferences.

4. A social welfare function f is utility-level invariant if it is invariant to arbitrary, but common, strictly increasing transformations ψ applies to every individual's utility function. Hence, f is permitted to depend only on the ordering of utilities both for and across individuals.

5. A social welfare function f is utility-difference invariant if it is invariant to strictly increasing transformations of the form $\psi^i(u^i) = a^i + bu^i$, where $b > 0$ is common to each individual's utility transformation. Hence, f is permitted to depend only on the ordering of utility differences both for and across individuals.

6. Assume that the set of social states X is non-singleton convex subset of Euclidean space and that all social choice functions f, under consideration satisfy strict welfarism (i.e., U, WP, IIA, and PI), where U means that f maps continuous individual utility functions into a continuous social utility function.

7. Consequently, we may summarize f with a strictly increasing continuous function $W : \mathbf{R}^N \to \mathbf{R}$ with the property that for every

continuous $\mathbf{u}(.) = (u^1(.), ..., u^N(.))$ and every pair of states x and y,

$$f_{\mathbf{u}}(x) \geq f_{\mathbf{u}}(y) \text{ iff } W(u^1(x), ..., u^N(x)) \geq W(u^1(y), ..., u^N(y)).$$

8. The extent to which utility is assumed to be measurable and interpersonally comparable can best be viewed as a question of how much information society uses when making social decision. For this purpose we impose two more requirement on the social welfare function.

9. Anonymity (A): Let $\overline{\mathbf{u}}$ be a utility N-vector, and let $\widetilde{\mathbf{u}}$ be another vector obtained from $\overline{\mathbf{u}}$ after some permutation of its elements. Then $W(\overline{\mathbf{u}}) = W(\widetilde{\mathbf{u}})$.

10. Hammond Equity (HE): Let $\overline{\mathbf{u}}$ and $\widetilde{\mathbf{u}}$ be two distinct utility N-vectors and suppose that $\overline{\mathbf{u}}^k = \widetilde{\mathbf{u}}^k$ for all k except i and j. If $\overline{u}^i < \widetilde{u}^i < \widetilde{u}^j < \overline{u}^j$, then $W(\widetilde{\mathbf{u}}) \geq W(\overline{\mathbf{u}})$.

11. Theorem (Rawlsian Social Welfare Functions). A strictly increasing and continuous social welfare function W satisfies HE iff it can take the Rawlsian form, $W = \min[u^1, ..., u^N]$. Moreover, W then satisfies A and is utility-level invariant.

12. Theorem (Utilitarian Social Welfare Functions). A strictly increasing and continuous social welfare function W satisfies A and utility-difference invariance iff it can take the utilitarian form, $W = \sum_{i=1}^{n} u^i$.

3 Justice

1. Economics and Philosophy have jointly considered the moral character that the social choice has to be made. Guidance has been sought by appeal to axiomatic theory of justice that accept the social welfare approach to social decision making. There are two historical traditions. One is the utilitarian tradition, associated with David Hume, Adam Smith, Jeremy Bentham, and John Mill. The other is the "contractarian" tradition, associated with John Locke, Rouseau, and Kant. More recently, these two traditions have been refined and articulated by the work of Harsanyi (1953, 1955, 1975) and Rawls (1971), respectively.

2. Both Harsanyi and Rawls accept the notion that a just criterion of social welfare must be one that a rational person would choose

if he were "fair-minded". To help ensure that the choice be a fair-minded, each imagines an "original position", behind what Rawls calls a "veil of ignorance", in which the individual contemplates this choice without knowing what his personal situation and circumstances in society actually will be. Thus, each imagines the kind of choice to be made as a choice under uncertainty over who you will end up having to be in the society you prescribe.

3. Harsanyi's approach. First, he accepted expected utility theory with uncertainty. A person's preferences can be represented by a VNM utility function over social states, $u^i(x)$, which is unique up to a positive affine transforms. By the principle of insufficient reason, he then suggests that a rational person in the original position must assign an equal probability to the prospect of being in any other person's shoes within the society. If there are N people in society, there is therefore a probability $1/N$ that i will end up in the circumstances of any other person j. Person i therefore must imagine those circumstances and imagine what his preference $u^j(x)$ would be. Because a person might end up with any of N possible "identities", a rational evaluation of social state x then would be made according to its expected utility

$$\sum_{i=1}^{n} \frac{1}{N} u^i(x).$$

4. In a social choice between x and y, the one with higher expected utility must be preferred. But this is equivalent to saying that x is socially preferred to y iff

$$\sum_{i=1}^{n} u^i(x) > \sum_{i=1}^{n} u^i(y),$$

a pure utilitarian criterion.

5. Rawls views the choice problem in the original position as one under complete ignorance. Assuming people are risk averse, he argues that in total ignorance, a rational person would order social states according to how he or she would view them were they to end up as society's worst-off member. Thus x will be preferred to y as

$$\min[u^1(x), ..., u^N(x)] > \min[u^1(y), ..., u^N(y)],$$

a purely maxmin criterion.

6. Rawls' maxmin criterion is a very special case of Harsanyi's utilitarianism, that is, the one when individuals are infinitely risk averse.

4 Social Choice and The Gibbard-Satterthwaite Theorem

1. We have focused on the task of aggregating the preferences of many individuals into a single preference relation for society. This task is a formidable one. Indeed, it cannot be carried out if we insist on all of Arrow's conditions.

2. How does society find out the preferences of its individual members? In addition to the problem of coherently aggregating individual rankings into a social ranking, there is the problem of finding out individual preferences in the first place.

3. Assume the set of social states, X, is finite and each of the N individuals in society is permitted to have any preference relation on X. That is, we assume unrestricted domain, U.

4. Any function $c(.)$ mapping all profiles of individual preferences on X into a choice from X, and whose range is all of X is called a social choice function.

5. A social choice function $c(.)$ is dictatorial if there is an individual i such that whenever $c(R^1, ..., R^N) = x$ it is the case that xR^iy for every $y \in X$.

6. A social choice function $c(.)$ is a strategy-proof when, for every individual, i, for every pair R^i and $\widetilde{R}^i$ of his preferences, and for every profile $\mathbf{R}^{-i}$ of others' preferences, if $c(R^i, \mathbf{R}^{-i}) = x$ and $c(\widetilde{R}^i, \mathbf{R}^{-i}) = y$, then xR^iy.

7. The Gibbard-Satterthwaite Theorem. If there are at least three social states, then every strategy-proof social choice function is dictatorial.

8. The message form the Gibbard-Satterthwaite Theorem is that, in a rich enough setting, it is impossible to design a non-dictatorial system in which social choice are made upon self-reported preferences without introducing the possibility that individual can gain by lying.

5 Exercises

1. Suppose that at a fixed input price $\overline{w}$ a firm produces output with the differentiable and strictly convex function $c(q, h)$, where q is the output level and h is the pollution generated by the firm. The pollution affects a consumer whose derived utility function is $u = \phi(h) + w$. The action of the firm and the consumer do not affect any market prices.

   (a) Derive the first-order condition for the firm's choice of q and h.

   (b) Derive the first-order conditions characterizing the Pareto optimal levels of q and h.

   (c) Suppose that the government taxes the firm's output level. Can this tax restore efficiency? Why or why not?

2. Suppose we have one firm and one consumer. The firm produces an output x which it sells in the competitive market. However, the production of x imposes a positive externality or a benefit $e(x)$ on the consumer. Let p be the price of the output.

   (a) Find out the social optimal production level and the profit maximization production level.

   (b) How can the government use subsidies to implement the social optimal production level?

3. Suppose Lake Yangcheng can be freely access by fishermen. The cost of sending out a boat on the lake is $c > 0$. When n boats are sent out onto the lake, $f(n)$ crabs are caught in total (so each boat catches $f(n)/n$ crabs), where $f(0) = 0$, $f'(n) > 0$ and $f''(n) < 0$ at all $n \geq 0$. The price of crab is $p > 0$, which is unaffected by the level of the catch from the lake.

   (a) Characterize the equilibrium number of boats that are sent out on the lake.

   (b) Characterize the optimal number of boats that should be sent out on the lake. Compare this answer with your answer to (a).

   (c) What per-boat crab tax would restore efficiency?

   (d) Suppose that the lake is instead owned by a single individual who can choose how many boats to send out. What level would this owner choose.